

Sina Abedini

Biochemistry Master's Student at McGill University

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Montreal, Canada

Personal Website

Research Interests

- Tissue Engineering
- Cancer bioengineering
- Genetic Engineering
- Tumor Microenvironment
- Protein Biochemistry
- Drug Delivery
- Host-Tumor Interactions
- Biomaterials
- Drug Discovery

Research Project

Fusion-Protein Strategy for Structural Studies of the Bacterial Magnesium Transporter CorB

Fusion of the Fab recognizable BRIL domain into target membrane protein for structural and functional studies

- **Molecular Biology Techniques:** Designed and cloned expression constructs using PCR and Gibson assembly; purified plasmid DNA and confirmed construct integrity.
- **Protein Biochemistry:** Expressed and purified recombinant membrane proteins; performed size-exclusion chromatography and SDS-PAGE to assess purity and integrity; incorporated transport proteins into liposomes for proteoliposome transport assays.
- **Cryo-EM Preparation:** Contributed to sample preparation for single-particle Cryo-EM analysis of membrane proteins, including nanodisc incorporation and Fab assisted fiducial mark strategies.
- **Critical Thinking and Troubleshooting:** Optimized protein expression and purification conditions, troubleshooting low-yield issues through iterative construct design.
- **Collaboration and Communication:** Worked within a multidisciplinary team; presented results in lab meetings and contributed to project planning.

Education

Master of Science (M.Sc.) in Biochemistry

School of Biomedical Sciences - Department of Biochemistry - McGill University

September 2023 – Present

Montreal, Canada

McGill University

- cGPA: 3.65/4

Bachelors of Science (B.S.) Degree in Biotechnology

School of Biology - University of Tehran

September 2019 – August 2023

Tehran, Iran

University of Tehran

Ranking

- Overall GPA: 18.69/20 (3.9/4)

High School Diploma in Natural Sciences

Mofid 2 High School

September 2016 – August 2019

Tehran, Iran

- Overall GPA: 19.65/20 (4/4)

Honors and Awards

- Best Poster Presentation Award at the 7th Annual CRBS symposium (2025)
- Grad Excellence Award in Biochemistry (2023)
- Awarded McGill University Full scholarship for graduate program in Biochemistry: tuition fully covered (2023)

- Ranked 2nd in the Biotechnology class of 2023, qualified for direct entry to fully funded masters program at the University of Tehran (2022)
- Awarded University of Tehran Full scholarship for undergraduate program: tuition fully covered by the government (2019)
- Ranked top 0.5 percent of 700000 attendees of the nationwide natural sciences matriculation exam (2019)
- Ranked 7th out of 139000 participants in the nationwide English language matriculation exam (2019)

Teaching Experience

Teaching Assistant

McGill University

📅 January 2026

📍 Department of Biochemistry

- **Biophysical Methods in Biochemistry**

👤 Dr. Martin Schmeing

Teaching Assistant

University of Tehran

📅 September 2022 - Dec 2022

📍 School of Biology

- **Biochemistry:** 👤 Dr.S.Safarian
 - Teaching the class topics from general and organic chemistry complementary to the biochemistry course
 - Reinforcing class lessons
 - Familiarizing the students with the course at hand
 - Answering student's questions
- **Specilized English texts:** 👤 Dr.V.Hassanzadeh
 - Familiarizing students with the overall structure and writing style of textbooks and papers written in English
 - Practicing reading, speaking and writing in academic and scientific settings
 - Reinforcing class lessons
 - Answering student's questions
- **General Biology:** 📍 School of Mathematics and Computer Science
👤 Dr.R.Naderloo
 - Teaching the students basic concepts of biology and genetics with emphasis on the central dogma of molecular biology
 - Relating biological concepts to data structure and paving the way into the field of Bioinformatics
 - Answering student's questions

Highschool Tutoring

Mofid 2 High school

📅 Sep 2022

- **Biology and Chemistry:**
 - Reviewing topics from first, second and final year of high school biology
 - Guidance counselling for preparation for the matriculation exam
 - Answering student's questions

Industry Experience

Internship at Quality Control Laboratory

Zist Darman Mahan Pharmaceutical Company

📅 August 2022

📍 Pardis Technology Park, Tehran, Iran

👤 biodep.ir

- Gained hands-on experience with standard operating procedures (SOPs) and Good Laboratory Practice (GLP) in a pharmaceutical QC environment
- Observed and assisted in microbial culture preparation, cell counting, and differential QC tests for probiotic products
- Learned and supported stages of pharmaceutical production, including fermentation, purification, and packaging processes
- Became familiar with quality control instruments and workflows used in probiotic drug development
- Developed a foundational understanding of regulatory compliance, documentation practices, and industry-level production cycles

Laboratory Skills

- Recombinant Protein Expression & Purification
- Gel Electrophoresis & DNA Gel extraction
- Molecular Cloning
- Size Exclusion Chromatography
- SDS-PAGE
- PCR
- Nanodisc preparation
- Proteoliposome preparation
- Plasmid purification
- Bacterial transformation and culture

Related Coursework

Graduate Coursework

Structural Pharmacology: (PHAR505) 3/4	Protein Biology and Proteomics: (BIOC605) 4/4
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Undergraduate Coursework

Nanobiotechnology: 18.5/20 (4/4 GPA)	Organic Chemistry: 18.75/20 (4/4 GPA)	Structural Biochemistry: 18.75/20 (4/4 GPA)
Metabolism Biochemistry: 19.5/20 (4/4 GPA)	Calculus 1: 19.9/20 (4/4 GPA)	Calculus 2: 17/20 (4/4 GPA)
Biostatistics: 16/20 (4/4 GPA)	Biochemical Engineering: 19.3/20 (4/4 GPA)	General Chemistry 2: 17.4/20 (4/4 GPA)
Molecular Genetics: 18.12/20 (4/4 GPA)	Principles of Genetic Engineering: 19.2/20 (4/4 GPA)	Cell and Molecular Biology: 15.75/20 (3/4 GPA)
Microbial Physiology: 20/20 (4/4 GPA)	Large Scale Production: 19.5/20 (4/4 GPA)	Immunology: 18.5/20 (4/4 GPA)
Virology: 20/20 (4/4 GPA)	Cell and Tissue Culture: 17.5/20 (4/4 GPA)	Medical Biotechnology: 19.5/20 (4/4 GPA)
Animal Biology and Physiology: 19.9/20 (4/4 GPA)	Computer Programming and Data Structure: 17.5/20 (4/4 GPA)	Basic Genetics: 18.8/20 (4/4 GPA)
Stem Cell Biotechnology: 20/20 (4/4 GPA)	Biomaterials and Tissue Engineering: 20/20 (4/4 GPA)	Bioinformatics: 18.5/20 (4/4 GPA)

Selected presentations

Zolgensma®

[Presentation on the structure and mechanism of action of the gene therapy drug Zolgensma used in the treatment of SMA patients](#)

📅 May 2023

📍 Human Genetics and Genetic Engineering Courses

🔗 [view](#)

CRISPR-Cas

[Presentation on the discovery and utilization of CRISPR-Cas technology](#)

📅 May 2022

📍 Virology Course

🔗 [view](#)

Mechanical Forces in Tissue Engineering

[Presentation on Mechanical Modulation of Cell Function in Tissue Engineering](#)

📅 Jan 2023

📍 Biomaterials and Tissue Engineering Course

🔗 [view](#)

Homologous Recombination

[Team presentation on DNA damage repair by mechanism of Homologous Recombination](#)

📅 Nov 2021

📍 Molecular Genetics Course

🔗 [view](#)

Vascular Endothelial Growth Factor

[Presentation on the biological role of Vascular Endothelial Growth factor \(VEGF\)](#)

📅 Jan 2021

📍 Molecular Genetics Course

🔗 view 📺

Microbiologically Influenced Corrosion

Presentation on the economic damage of corrosion and the role of microorganisms in induction or acceleration of corrosion

📅 Jul 2022

📍 Microbial Biotechnology Course

Language Skills

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| <ul style="list-style-type: none">• English:<ul style="list-style-type: none">- TOEFL: 113
(Reading: 30, Listening: 29, Speaking: 30, Writing: 24) | <ul style="list-style-type: none">- GRE: 312
(Verbal: 155, 66th percentile, Quantitative: 157, 57th percentile) |
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Computer skills

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| <ul style="list-style-type: none">• Microsoft Suit:
Powerpoint, Word, Excel | <ul style="list-style-type: none">• Programming:
Python | <ul style="list-style-type: none">• Analysis Tools:
SPSS, PyMOL, SnapGene, AlphaFold |
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References

- **Dr. Kalle Gehring (Supervisor)**, Professor, Department of Biochemistry, Canada Research Chair (Tier 1) in Structural Studies of Neurodegenerative Diseases, Associate member, Department of Physics, McGill University, Email: ✉ kalle.gehring@mcgill.ca
🌐 Lab Website
- **Dr. Albert Berghuis**, Professor, Department of Biochemistry, McGill University, Email: ✉ albert.berghuis@mcgill.ca